

A Small-Scale Reproduction of Semi-Manual Evaluation for Italian Grammatical Error Correction for MultiGEC-2025

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Abstract

Automatic metrics for Grammatical Error Correction (GEC) often fail to capture fine-grained linguistic quality, particularly for systems that produce fluent yet minimally edited outputs. Östling et al. (2024) proposed a semi-manual evaluation framework combining human judgments of grammaticality, fluency and meaning preservation as an alternative to purely automatic metrics. This paper presents a small-scale reproduction of that methodology for Italian GEC, a language underrepresented in GEC evaluation research. We evaluate two systems, a MultiGEC baseline and OmniGEC, on a subset of the MERLIN Italian learner corpus using the the same annotator Tkinter tool used in the Shared Task 2025. Our results show that the semi-manual evaluation method works well for Italian and that it helps to see the real differences between the two used models that automatic metrics, in general, usually struggle to capture, pointing out Italian-specific errors, like mistakes with articles and prepositions.

1. Introduction

Grammatical Error Correction (GEC) is a core task in Natural Language Processing with applications in language learning, writing assistance and professional communication. While recent neural models achieve high performance according to automatic metrics, there is increasing concern that such metrics fail to reflect true linguistic quality, especially for systems producing minimal edits or stylistic paraphrases.

This issue is particularly salient for languages other than English. Italian, despite its rich morphology and syntactic complexity, remains underrepresented in GEC evaluation research. The MultiGEC-2025 shared task [4][5] addressed this gap by providing multilingual evaluation benchmarks, but evaluation still relies heavily on automatic scores.

Östling et al. (2024) [1] proposed a semi-manual evaluation methodology that integrates structured human judgments, aiming to better assess “really good” GEC systems.

The goal of this paper is to reproduce the semi-manual evaluation methodology on a small scale for the Italian language. We focus on three aspects: first, methodologically, we investigate whether the framework can be effectively applied to Italian GEC; second, we make a performance system comparison between a MultiGEC baseline model and OmniGEC [6] model using this semi-manual evaluation; and third, we perform a linguistic

analysis, looking which Italian-specific error types are better or worse handled by current GEC systems.

2. Related Work

Early GEC evaluation relied primarily on reference-based metrics such as precision, recall and F-score, often implemented via ERRANT. However, such metrics struggle with alternative correct corrections, over-penalize stylistic variation and inadequately reflect fluency and acceptability.

The MultiGEC shared tasks introduced multilingual GEC evaluation, but still largely depend on automatic scoring. In response, Östling et al. (2024) argue that modern GEC systems require human-centered evaluation, proposing a semi-manual framework combining system outputs, human corrections and structured quality ratings. OmniGEC represents a recent multilingual neural approach designed to generalize across languages, including Italian.

3. Dataset

For this study, we used the Italian portion of the MERLIN learner corpus (Wisniewski et al., 2013; Boyd et al., 2014) [2], as distributed in the MultiGEC-2025 shared task. MERLIN is a trilingual, error-annotated written learner corpus for German, Italian and Czech, created within the MERLIN project (2012–2014). It contains essays written by learners of these languages as a second language (L2), collected from standardized language tests and aligned with the Common European Framework of Reference for Languages (CEFR). Each Italian essay is accompanied by minimal target hypotheses (TH1) that correct orthographic and grammatical errors and, in some cases, extended target hypotheses (TH2) addressing sociolinguistic or pragmatic deviations; for the purposes of this study, only the minimal hypotheses (TH1) were used.

We focused on a subset of the development split, consisting of 48 sentences, which allows for a small-scale manual evaluation while remaining methodologically sound. The development split was chosen because it is cleaner and provides sentences that are aligned with minimal target hypotheses, making them suitable for fine-grained semi-manual analysis.

Initially, the essays contained multiple sentences per essay. We attempted to automatically split the essays into individual sentences using a Python script to match the original sentences to the reference minimal hypotheses (from `it-merlin-orig-test.md` and `it-merlin-refl-dev.md`). However, automatic splitting caused alignment issues, so we manually segmented the essays into individual sentences, resulting in a total of 48 single-sentence essays ready for evaluation.

Each sentence was structured with an essay identifier and the original text, for example (from `it-merlin-orig-dev_simplified.md`):

essay_id = 1365_0100191_01_1

Egregio Signore, sto scrivendo per annunciare la mia intenzione di frequentare uno di loro corsi rilevanti, cioè il corso estivo al mare.

essay_id = 1365_0100191_01_2

Sono un insegnante quindi fortunatamente ho vacanze di due mesi dal Luglio al Agosto.

essay_id = 1365_0100191_01_3

Penso che questo corso sia il migliore per me tanto sarò libero in questi mesi e mi piace essere in un paese mediterraneo d'estate.

This sentence-level structure allows both the baseline system and OmniGEC to process the data and facilitates detailed human judgment along the three core dimensions of our semi-manual evaluation: grammaticality, fluency and meaning preservation.

The MERLIN corpus is openly available under a CC BY-SA 4.0 license via CLARIN (<http://hdl.handle.net/20.500.12124/6>), ensuring transparency and reproducibility of our evaluation.

4. Methodology

4.1 Systems

We evaluated two GEC systems on the Italian development subset of the MERLIN corpus:

- Baseline System (Italian Mistral): A simplified version of the baseline system provided in the MultiGEC-2025 repository (<https://github.com/spraakbanken/multigec-2025>), modified to run on the GU University GPU server. Due to memory constraints, the model operates in 4-bit quantized inference mode using bitsandbytes.
- OmniGEC (Italian): A multilingual GEC system evaluated in minimal mode (<https://huggingface.co/lang-uk/OmniGEC-Minimal-8B>), also simplified to run on the GPU server with 4-bit quantization.

Attempts to use other models from the MultiGEC-2025 shared task were not pursued, as they were either too computationally demanding to run on our GPU setup (e.g. the winner of the MultiGEC-2025 shared task UAM-CSI model [3]) or specifically designed for Swedish, making them unsuitable for Italian. Consequently, our study relied on those two above-described systems. Running them required substantial adaptation. Both models were initially incompatible with the simplified markdown input files (it-merlin-orig-dev_simplified.md) and produced incorrect outputs. We worked extensively on prompt engineering to ensure minimal corrections while preserving meaning. All outputs were cleaned and aligned to generate Markdown files (baseline_output.md and omnigec_output.md) suitable for subsequent evaluation.

4.2 Semi-Manual Evaluation Framework

We adopted the semi-manual evaluation framework proposed by Östling et al. (2024) using the Tkinter-based annotation GUI (annotator.py) provided in the gec-evaluation repository (<https://github.com/robertostling/gec-evaluation>) to manually annotate each sentence and subsequently analyse the results with analyze_annotations.py. We followed the original semi-manual evaluation methodology annotating as following:

1. Grammaticality: how well-formed the sentence is, independently of the reference.
2. Fluency: how natural the sentence sounds to a native speaker.
3. Meaning Preservation: how closely the system output preserves the intended meaning, judged against the reference sentence.

For each sentence, the annotator is presented with:

- The system-generated correction (from either the baseline or OmniGEC model)
- The reference minimal target hypothesis (TH1)

The annotation procedure is as follows:

1. The annotator first evaluates grammaticality and fluency without consulting the reference.
2. The Tkinter GUI allows the annotator to edit the system outputs minimally to correct errors.
3. Each dimension is rated on a 1–4 scale.
4. After saving, the reference sentence is revealed and the annotator assigns a meaning preservation rating.
5. All annotations are confirmed and stored in JSON format for later analysis.

This procedure ensures a systematic and replicable evaluation, capturing both surface-level grammatical quality and semantic fidelity. It also allows identification of Italian-specific error patterns that may not be detected by automatic metrics alone. Finally, we analysed the results with analyze_annotations.py.

In Table 1, we can see an example of the described semi-manual annotation system.

Version	Text	G	F	M
Student	Si vuole studiare lingue straniere qui può imperare.	-	-	-
Reference (TH1)	Se si vuole studiare lingue straniere qui si può imparare.	-	-	-
Baseline output	Si vuole studiare lingue straniere qui.	1	1	1
Baseline + post-edit	Qui si possono studiare lingue straniere.	4	4	1
OmniGEC output	Si vuole studiare lingue straniere qui, può imperare.	1	1	1
OmniGEC + post-edit	Se si vuole studiare lingue straniere, qui si può imparare.	4	4	4

Table 1: Scores for (G)rammatical, (F)luency and (M)eaning preservation for selected examples from the annotated data.

The baseline system deletes the conditional construction entirely, producing a short and grammatical sentence that no longer preserves the intended meaning (M1), despite being correctable to a fluent sentence through post-editing. OmniGEC retains more of the original structure but fails to correct the critical lexical error (*imperare* vs. *imparare*), resulting in low grammaticality and fluency scores. However, after minimal post-editing, the OmniGEC output can be brought fully in line with the reference, illustrating how the framework separates surface well-formedness from semantic fidelity and edit distance.

5. Results

We manually annotated 48 sentences from the Italian learner corpus using the `annotator.py` Tkinter GUI, following the semi-manual evaluation procedure. Each sentence was scored for Grammaticality (G), Fluency (F) and Meaning Preservation (M) on the scales defined in Section 4.2.

5.1 Quantitative Analysis

Model	Grammaticality	Fluency	Meaning Preservation	NLD
Baseline	3.3	3.3	3.4	0.110
OmniGEC	3.4	3.4	3.8	0.064

Table 2: Human evaluation: mean score per system and assessment dimension. Higher is better for all assessments (range: 1–4), while lower is better for NLD.

where NLD is the normalized Levenshtein distance between the system output and the human reference, averaged across sentences. Lower values indicate outputs closer to the reference.

Each dimension is rated on a 1–4 scale. Grammaticality assesses the well-formedness of the system output independently of the reference sentence. It ranges from Perfect, where the sentence is fully grammatical with at most minor typographical or collocational issues, to Comprehensible, which allows minor grammatical errors that do not obscure meaning. Somewhat comprehensible sentences contain more serious errors that introduce ambiguity or multiple plausible interpretations, while Incomprehensible sentences are so ill-formed that they are difficult to understand or correct.

Fluency evaluates how natural the sentence sounds to a native speaker, regardless of whether it preserves the intended meaning. Sentences are rated from Extremely natural, if they could plausibly be produced by a native speaker in some context, to Extremely unnatural, if they sound clearly non-native or awkward due to word order, collocation or morphosyntactic issues. Low grammaticality implies low fluency, but grammatically correct sentences may still receive low fluency scores if they sound unnatural.

Meaning preservation measures how closely the system output matches the intended meaning of the human reference sentence. Unlike grammaticality and fluency, this parameter is judged relative to the reference. Outputs are classified as Identical when no meaningful semantic differences are present, Minor differences when small changes such as definiteness or number occur, Moderate differences when important lexical or relational meaning changes are introduced and Substantially different when the sentence conveys a clearly different message. The reference is assumed to reflect the correct intended meaning of the original learner sentence.

Both models generally produced grammatical sentences, with OmniGEC slightly outperforming the baseline (3.4 vs 3.3), particularly on sentences with complex tense and agreement constructions. For example, the baseline output “Ma nel futuro, come ti ho detto, vorrei lavorare all’estero e quindi devo sicuramente utilizzare le lingue straniere” was rated G3, whereas OmniGEC produced “Ma nel futuro - come Le ho detto - vorrei lavorare all’estero e ci naturalmente devo usare le lingue straniere” (G2). While OmniGEC occasionally introduces slightly awkward constructions, it largely preserves grammatical correctness. In terms of fluency, OmniGEC also scores higher (3.4) than the baseline (3.3), reflecting more natural phrasing, improved word order and smoother transitions. For instance, the baseline sentence “Spero che con le lingue straniere posso lavorare all’estero, posso incontrare e conoscere persone straniere, posso parlare a molta gente nel mondo” (F4) is outperformed by OmniGEC’s “Spero di poter lavorare all’estero con le lingue straniere, di poter incontrare e conoscere persone straniere, di poter parlare con molta gente nel mondo” (F3), which employs more standard Italian modal constructions such as “di poter lavorare”. OmniGEC also demonstrates clear gains in meaning preservation (3.8 vs 3.4), producing sentences that better align with the intended reference meaning. For example, the baseline “Sicuro ora devo studiare le lingue straniere perché ne necessito per l’università” (M1) is improved by OmniGEC to “Certo, adesso devo studiare le lingue straniere perché ho bisogno dei punti per l’università” (M3), correcting misinterpretations and better reflecting the target message. Error analysis shows that the baseline frequently produces literal, word-by-word translations with minor grammatical mistakes, while OmniGEC sometimes generates slightly unnatural phrasing, but preserves meaning more reliably and reduces pairwise normalized Levenshtein distance (NLD). Using multidimensional scaling on NLD, we visualized the relationship between original sentences, system outputs and human corrections (see Figure 1). The plot shows that OmniGEC tends to produce corrections closer to human edits, suggesting more minimal and conservative changes. It shows also that both systems move outputs closer to the original than to the reference.

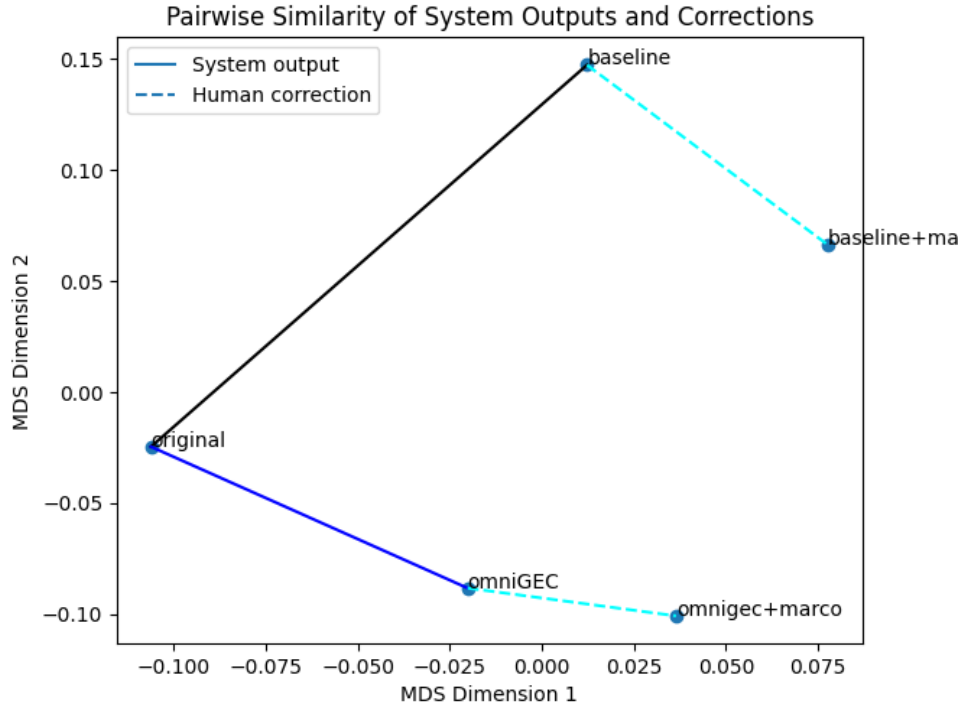


Figure 1: Multidimensional scaling (MDS) visualization of all text versions, based on normalized Levenshtein distance. The graph forms a tree rooted at the original learner sentences. Edges indicate transformations applied either by the systems (solid lines) or by the human annotator, the author (dot lines). Unlabeled leaf nodes correspond to human post-edits aimed at achieving full grammaticality, fluency and meaning preservation.

According to our results, the baseline achieves moderate scores in all dimensions (Grammaticality: 3.3, Fluency: 3.3, Meaning Preservation: 3.4) with a normalized Levenshtein distance (NLD) of 0.110. OmniGEC shows a consistent improvement across all metrics (Grammaticality: 3.4, Fluency: 3.4, Meaning Preservation: 3.8) and a lower NLD of 0.064, indicating closer alignment with human references and smoother, more accurate corrections. Low-scoring cases (G1-F1-M1) are typically short or idiomatic sentences that remain challenging for both models. Pairwise comparisons using `analyze_annotations.py --action compare` confirm that OmniGEC consistently outperforms the baseline on meaning preservation while achieving slightly higher grammaticality and fluency, producing corrections closer to the reference in both structure and intended meaning.

5.2 Qualitative Analysis

Manual inspection reveals that OmniGEC systematically outperforms the baseline in areas such as article insertion (e.g. “la città di X”), agreement corrections and avoiding unnecessary paraphrases. However, it sometimes over-simplifies sentences and under-corrects stylistic issues. Both systems share common challenges, including preposition choice (e.g. “a” vs. “di”) and capitalization of proper nouns. These patterns highlight the importance of conducting language-specific evaluation to capture nuanced errors in Italian.

6. Discussion

Our findings support the applicability of the semi-manual evaluation framework to Italian, demonstrating that it can capture nuanced differences between systems that automatic metrics often overlook. While both the baseline and OmniGEC produce generally grammatical sentences, the semi-manual annotation revealed clear improvements in OmniGEC’s handling of subject-verb agreement, word order and idiomatic expressions, producing outputs closer to the reference meaning. For example, in the sentence “Spero che con le lingue straniere posso lavorare all’estero, posso incontrare e conoscere persone straniere, posso parlare a molta gente nel mondo”, the baseline output preserves the words, but uses nonstandard modal construction. OmniGEC corrects this to “Spero di poter lavorare all’estero con le lingue straniere, di poter incontrare e conoscere persone straniere, di poter parlare con molta gente nel mondo”, improving naturalness and fluency while keeping the meaning intact.

Another example illustrates OmniGEC’s ability to align with formal usage and idiomatic phrasing: the baseline “Ma nel futuro, come ti ho detto, vorrei lavorare all’estero e quindi devo sicuramente utilizzare le lingue straniere” scores G3-F3-M2, whereas OmniGEC outputs “Ma nel futuro - come Le ho detto - vorrei lavorare all’estero e lì naturalmente dovrei usare le lingue straniere” (G2-F3-M4). Here, the model correctly adjusts the pronoun from informal *ti* to formal *Le* and refines the word order, producing a sentence closer to the reference.

A third example shows OmniGEC’s improvements in meaning preservation on short, context-sensitive sentences. For “Sicuro ora devo studiare le lingue straniere perché ne necessito per l’università”, the baseline preserves the literal content, but misrepresents the intended nuance (M1). OmniGEC outputs “Certo, adesso devo studiare le lingue straniere perché ho bisogno dei punti per l’università” (M3), clarifying the meaning while maintaining grammatical correctness and fluency.

These examples illustrate that OmniGEC reduces errors related to literal translations and awkward constructions, produces more standard Italian modal verbs and clitic placement and better handles idiomatic expressions. Pairwise comparisons confirm that OmniGEC consistently outperforms the baseline in meaning preservation while achieving slightly higher grammaticality and fluency, particularly in sentences with complex agreement or multi-clause structures. At the same time, certain short or idiomatic sentences remain challenging, underscoring persistent Italian-specific GEC difficulties and the value of a semi-manual, language-aware evaluation framework.

7. Limitations and Challenges

The study has several limitations. Only a single annotator performed the evaluation, introducing potential bias and the dataset size was relatively small, limiting generalizability. Certain sentence types, such as short exclamations, idiomatic expressions or highly context-dependent sentences, were particularly challenging for both systems, reflecting inherent limitations in GEC models for Italian. Despite these constraints, the study succeeds in demonstrating the feasibility of the semi-manual evaluation framework and provides interpretable, sentence-level insights.

8. Conclusion and Future Work

This paper presented a small-scale reproduction of the semi-manual GEC evaluation framework proposed by Östling et al. (2024), applied to Italian. The methodology transferred well, producing interpretable results that revealed both general trends and Italian-specific challenges in error correction. OmniGEC consistently outperformed the baseline in meaning preservation, with slight improvements in grammaticality and fluency, particularly on sentences requiring complex agreement, idiomatic phrasing or natural word order. Future work includes adding multiple annotators to reduce bias, expanding sentence coverage to increase robustness and systematically comparing semi-manual evaluation outcomes with fully automatic metrics to better understand their respective strengths and limitations.

References

1. Östling, R., et al. (2024). Evaluation of really good grammatical error correction.
2. Boyd, A., et al. (2025). The MERLIN corpus: Learner language and the CEFR.
3. Staruch, R. (2025). UAM-CSI at MultiGEC-2025: Parameter-efficient LLM fine-tuning for multilingual grammatical error correction.
4. Masciolini, A., et al. (2025). An overview of grammatical error correction for the twelve MultiGEC-2025 languages.
5. Masciolini, A., et al. (2025). Towards better language representation in natural language processing: A multilingual dataset for text-level grammatical error correction.
6. Kovalchuk, R., et al. (2025). Introducing OmniGEC: A silver multilingual dataset for grammatical error correction. Ukrainian Catholic University, Softserve, Lviv, Ukraine.